K Statistical sampling

Name: ______________________

Class: ______________________

Date: ______________________

Time: **37 minutes**

Marks: **32 marks**

Comments:

Q1.

Kevin is the Principal of a college.

He wishes to investigate types of transport used by students to travel to college.

There are 3200 students in the college and Kevin decides to survey 60 of them.

Describe how he could obtain a simple random sample of size 60 from the 3200 students.

(Total 4 marks)

Q2.

In a region of England, the government decides to use an advertising campaign to encourage people to eat more healthily.

Before the campaign, the mean consumption of chocolate per person per week was known to be 66.5 g, with a standard deviation of 21.2 g

(a) After the campaign, the first 750 available people from this region were surveyed to find out their average consumption of chocolate.

(i) State the sampling method used to collect the survey.

(1)

(ii) Explain why this sample should not be used to conduct a hypothesis test.

(1)

(b) A second sample of 750 people revealed that the mean consumption of chocolate per person per week was 65.4 g

Investigate, at the 10% level of significance, whether the advertising campaign has decreased the mean consumption of chocolate per person per week.

Assume that an appropriate sampling method was used and that the consumption of chocolate is normally distributed with an unchanged standard deviation.

(6)

(Total 8 marks)

Q3.

Neesha wants to open an Indian restaurant in her town.

Her cousin, Ranji, has an Indian restaurant in a neighbouring town. To help Neesha plan her menu, she wants to investigate the dishes chosen by a sample of Ranji's customers.

Ranji has a list of the 750 customers who dined at his restaurant during the past month and the dish that each customer chose.

Describe how Neesha could obtain a simple random sample of size 50 from Ranji's customers.

(Total 4 marks)

Q4.

Ellie, a statistics student, read a newspaper article that stated that 20 per cent of students eat at least five portions of fruit and vegetables every day.

Ellie suggests that the number of people who eat at least five portions of fruit and vegetables every day, in a sample of size n, can be modelled by the binomial distribution B(n , 0.20).

(a) There are 10 students in Ellie's statistics class.

Using the distributional model suggested by Ellie, find the probability that, of the students in her class:

(i) two or fewer eat at least five portions of fruit and vegetables every day;

(1)

(ii) at least one but fewer than four eat at least five portions of fruit and vegetables every day.

(2)

(b) Ellie's teacher, Declan, believes that more than 20 per cent of students eat at least five portions of fruit and vegetables every day. Declan asks the 25 students in his other statistics classes and 8 of them say that they eat at least five portions of fruit and vegetables every day.

(i) Name the sampling method used by Declan.

(1)

(ii) Describe **one** weakness of this sampling method.

(1)

(iii) Assuming that these 25 students may be considered to be a random sample, carry out a hypothesis test at the 5% significance level to investigate whether Declan's belief is supported by this evidence.

(6)

(Total 11 marks)

Q5.

Edna wishes to investigate the energy intake from eating out at restaurants for the households in her village.

She wants a sample of 100 households.

She has a list of all 2065 households in the village.

Ralph suggests this selection method.

> "Number the households 0000 to 2064. Obtain 100 different four-digit random numbers between 0000 and 2064 and select the corresponding households for inclusion in the investigation."

(a) What is the population for this investigation?

Circle your answer.

Edna and Ralph	The 2065 households in the village	The energy intake for the village from eating out	The 100 households selected

(1)

(b) What is the sampling method suggested by Ralph?

Circle your answer.

Opportunity	Random number	Continuous random variable	Simple random

(1)
(Total 2 marks)

Q6.

Explain the difference between a parameter and a statistic.

(Total 3 marks)